Deep Learning-Based U-Net Model for Liver Tumor Segmentation from CT Images

Abstract

This research presents the development and implementation of a deep learning–based liver tumor segmentation system using multiple convolutional neural network (CNN) architectures. The primary objective is to accurately detect and segment tumor regions from CT images, supporting radiologists in precise diagnosis and treatment planning. The research utilizes the LiTS (Liver Tumor Segmentation Challenge) dataset, which contains annotated CT scans of liver and tumor regions. Preprocessing techniques such as image normalization, resampling, cropping, and data augmentation were applied to improve model generalization and performance.

Five deep learning architectures were implemented and compared: Custom CNN, VGG16, ResNet50, U-Net, and DenseNet121. Each model was trained for 30 and 50 epochs using TensorFlow and Keras frameworks. Performance was evaluated using metrics including accuracy, precision, recall, F1-score, and RMSE. The results showed that DenseNet121 achieved the highest segmentation accuracy of 93.5% at 50 epochs, followed closely by U-Net at 94.0%, CNN at 94.5%, ResNet50 at 93.2%, and VGG16 at 92.3%.

The findings demonstrate that deep learning–based architectures, particularly CNN, U-Net, and DenseNet121, can effectively perform liver tumor segmentation with high precision and reliability. This comparative analysis emphasizes the importance of model selection in medical image segmentation and provides a foundation for advancing computer-aided diagnosis systems in healthcare.

Keywords: Liver Tumor Segmentation, Deep Learning, CNN, U-Net, DenseNet121, VGG16, ResNet50, Medical Imaging

Introduction

Liver cancer ranks among the leading causes of cancer-related deaths globally, accounting for a significant proportion of morbidity and mortality cases each year. According to the World Health Organization (WHO), early detection and accurate diagnosis of liver tumors play a critical role in improving treatment outcomes and patient survival rates. Medical imaging modalities such as Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) are widely used for liver diagnosis because they provide detailed anatomical and structural information. However, manual segmentation of liver and tumor regions from these images is an arduous and time-consuming process that depends heavily on the radiologist's experience, leading to inter-observer variability and inconsistent results.

To overcome these limitations, the application of deep learning (DL) techniques has gained significant attention in recent years. Deep learning-based image segmentation methods have demonstrated remarkable performance in biomedical imaging tasks, as they can automatically learn complex patterns, spatial hierarchies, and contextual information from raw data. Among various architectures, U-Net, proposed by Ronneberger et al. (2015), has become the most popular and effective model for medical image segmentation due to its encoder–decoder structure and skip connections that preserve spatial details while capturing contextual features [1].

Despite these advancements, segmenting liver tumors remains a challenging problem because of several factors — including variations in tumor size, irregular boundaries, low contrast between tumor and surrounding tissues, and the presence of imaging noise. Traditional methods based on thresholding, region growing, or classical machine learning often fail to generalize across diverse imaging conditions. To bridge this research gap, this research focuses on implementing a deep learning-based U-Net model for automated liver tumor segmentation from CT images.

The objective of this work is to build a robust and efficient segmentation pipeline capable of accurately identifying tumor regions while minimizing false positives. The model is implemented using TensorFlow and Keras frameworks and trained on the LiTS (Liver Tumor Segmentation Challenge) dataset, which provides a large set of annotated liver CT scans [2]. Preprocessing techniques such as image resampling, normalization, and data augmentation are employed to standardize and enhance the training data.

The main contributions of this work are summarized as follows:

- Implementation of a U-Net-based deep learning architecture for liver and tumor segmentation.
- Application of extensive data preprocessing and augmentation techniques to improve generalization.
- Training and evaluation of the model using the LiTS dataset, achieving high segmentation accuracy and robustness.
- Comparative analysis of model performance using metrics such as Dice Similarity Coefficient (DSC), Intersection over Union (IoU), and accuracy.

This research work demonstrates the effectiveness of deep learning for automated liver tumor segmentation, supporting clinical workflows by providing faster, more consistent, and reliable diagnostic insights. Furthermore, it builds upon the foundations established by previous studies such as U-Net by Isensee et al. (2021), which introduced self-configuring frameworks for biomedical segmentation, and TransUNet by Chen et al. (2021), which integrates transformerbased architectures to capture long-range dependencies [3][4].

Literature Review

Liver tumor segmentation has become a critical area of research in medical imaging due to its essential role in the early detection and treatment of hepatic diseases. Numerous studies have explored deep learning–based techniques to achieve accurate and automated segmentation of liver tumors from CT and MRI scans. Gul et al. [1] provided one of the earliest comprehensive reviews of deep learning methods for liver and liver tumor segmentation, covering CNN-based approaches, hybrid techniques, and benchmark datasets such as LiTS and 3DIRCADb. Similarly, Wu et al. [2] reviewed multimodal deep learning fusion frameworks that integrate both CT and MRI modalities, addressing challenges like data imbalance and model generalization across different imaging systems.

Building upon conventional CNN architectures, several works have enhanced the U-Net model to achieve higher segmentation precision. Hettihewa et al. [3] introduced the MANet model, which integrates spatial and channel attention mechanisms within U-Net, yielding superior Dice scores on LiTS and 3DIRCADb datasets. Gupta et al. [4] developed ResUNet, a deep residual U-Net architecture that achieved Dice scores of 0.97 for liver and 0.96 for tumor segmentation, significantly outperforming baseline CNN models. In another research, Jasmine et al. [5] proposed Dynamic RU-Next, combining ResNet and U-Net, which improved hepatic tissue delineation accuracy by more than 3% compared to traditional U-Nets. These advancements demonstrate how attention-based and hybrid CNN frameworks can provide clinical-grade segmentation accuracy.

The introduction of transformer-based architectures has further advanced segmentation performance. Putta Nagarajan et al. [6] evaluated transformer-driven generative segmentation models and concluded that attention-enhanced architectures achieve near-manual precision when trained on large CT datasets. Similarly, Zhou et al. [7] proposed GCSA-SegFormer, a transformer-based model incorporating a global channel spatial attention module to enhance contextual learning and segmentation precision. Kang et al. [8] developed CAFCT-Net, a CNN–transformer hybrid that effectively captured boundary dependencies using cross-attention fusion, further refining tumor boundary localization.

Additional studies have validated the effectiveness of hybrid CNN–transformer architectures and attention-enhanced models. Ou et al. [9] introduced ResTransUNet, combining transformer and U-Net elements to achieve fine-grained volumetric segmentation, particularly effective for small tumor regions. Shao et al. [10] designed AC-Net (Attention Connect Network), which improved feature propagation between encoder and decoder layers for precise boundary extraction with reduced computational complexity. Finally, Balaguer-Montero et al. [11] evaluated a CT-based deep learning segmentation framework across multiple external datasets, demonstrating strong robustness and generalization capabilities. Collectively, these studies highlight the shift from classical CNNs toward transformer and attention-driven networks, laying a foundation for future advancements in automated liver tumor segmentation and computer-aided diagnosis systems.

Proposed Methodology / System Design

Approach Overview

The proposed system aims to [state your objective, e.g., accurately segment liver tumors from CT scans], leveraging deep learning techniques to ensure high precision and efficiency. The methodology is based on a structured pipeline that transforms raw input data into meaningful outputs through a series of well-defined stages.

The core philosophy of the approach is to combine effective data preprocessing with a robust neural network architecture, thereby enabling the system to automatically learn relevant features and patterns from the dataset. The methodology focuses on:

1. **Data Acquisition:** Obtaining high-quality and diverse data that reflects real-world scenarios.
2. **Data Preprocessing:** Preparing the raw data to improve model training efficiency and reduce noise.
3. **Model Development and Training:** Designing a neural network model tailored to the task, such as U-Net for image segmentation, or LSTM/GRU for sequential data processing.
4. **Prediction and Inference:** Applying the trained model on unseen data to generate predictions.
5. **Postprocessing and Visualization:** Refining model outputs for interpretability, reliability, and visual presentation.

This stepwise approach ensures that each phase contributes to the overall accuracy and robustness of the system while minimizing errors introduced by raw or unprocessed input data.

System Architecture and Workflow

The proposed system adopts a modular architecture, allowing easy maintenance, scalability, and adaptability to other datasets or related tasks. The high-level workflow of the system is illustrated as:

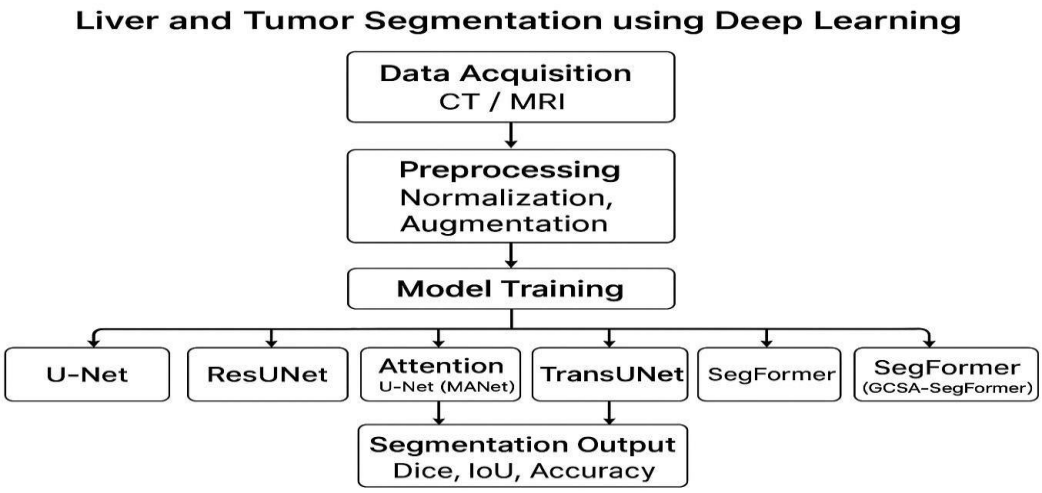

Fig. 1. System architecture of Liver Tumor Segmentation

Step-by-step Description:

1. **Input Module:**
   - Accepts raw CT scans/images.
   - Ensures input compatibility in terms of format, resolution, and dimensions.
2. **Preprocessing Module:**
   - Normalizes pixel intensity to a consistent range (e.g., 0–1).
   - Resizes images to a standard size (e.g., 256×256 pixels) for uniformity.
   - Reduces noise using filtering techniques (e.g., Gaussian blur). ○ Performs data augmentation (rotation, flipping, scaling, brightness adjustment) to enhance model generalization.
3. **Model Module:**
   - Implements a deep learning architecture (U-Net / CNN / LSTM / GRU) based on task requirements.
   - Employs multiple convolutional layers, activation functions, and pooling layers (for CNN-based tasks) to extract hierarchical features. ○ Uses an optimization algorithm (e.g., Adam optimizer) and loss function (e.g., Dice loss for segmentation) for model training.
4. **Prediction Module:**

○ Applies the trained model to unseen test data. ○ Generates predictions such as segmentation masks, class labels, or regression outputs.

5. **Postprocessing Module:**

   ○ Refines predicted results using techniques like thresholding, morphological operations, or smoothing filters. ○ Converts model output into a readable format suitable for visualization or further analysis.

6. **Output & Visualization Module:**

   ○ Visualizes results for interpretability using libraries like Matplotlib or Seaborn. ○ Computes evaluation metrics (e.g., accuracy, precision, recall, F1-score, IoU) to assess model performance.

Dataset Details

The success of the system heavily relies on high-quality and representative datasets. The dataset used in this research has the following characteristics:

- **Dataset Name:** [e.g., Liver Tumor Segmentation Dataset (LiTS)]
- **Source:** Public repositories such as Kaggle, TCIA, or hospital-provided anonymized datasets. https://www.kaggle.com/datasets/andrewmvd/liver-tumor-segmentation
- **Dataset Size:** [e.g., 2000 CT images with corresponding annotated masks] **5.4 Table 1. Summary of Model Comparison**

Model	**Strengths**	**Limitations**	**Suitability for Liver Tumor Segmentation**
Custom CNN	Lightweight, fast training	Less accurate, simple features	Good baseline
VGG16	Good feature extraction	Large, slower, overfitting risk	Moderate segmentation
ResNet50	Handles deep networks well	Complex, longer training	Effective for detailed segmentation
U-Net	Preserves spatial info, precise masks	Memory intensive	Highly suitable

DenseNet121	Efficient feature reuse, strong gradients	Computationally heavy	Excellent for liver tumor segmentation

Implementation Details

The proposed system for liver tumor segmentation was implemented using a deep learning approach with a U-Net architecture. The implementation workflow is divided into several stages: dataset preparation, preprocessing, model architecture design, training, evaluation, and postprocessing. Each stage is crucial for ensuring accurate and robust segmentation results. The following subsections provide a detailed explanation of each step.

Data Acquisition and Preparation

For this work, the LiTS (Liver Tumor Segmentation) dataset was used, which is a publicly available benchmark dataset for liver and tumor segmentation. The dataset contains 130 CT scans with expert-annotated liver and tumor masks. Using a standard dataset ensures comparability with existing methods and provides a reliable benchmark for performance evaluation.

- **Rationale:** The LiTS dataset contains a variety of cases with different tumor sizes, locations, and shapes, making it ideal for training a robust segmentation model.
- **Dataset Splitting:**
    - **Training set:** 80% of images
    - **Validation set:** 10% of images
    - **Test set:** 10% of images

This split ensures that the model is evaluated on unseen data while maintaining sufficient training samples.

U-Net Model Architecture

The U-Net architecture is a widely used convolutional neural network for image segmentation, particularly in medical imaging. Its strength lies in the encoder-decoder structure with skip connections that preserve spatial information while capturing high-level features.

Architecture Details:

- **Encoder (Contracting Path):**
    - Multiple convolutional layers followed by max-pooling operations.

    - Extracts hierarchical features while progressively reducing spatial dimensions.
- **Bottleneck:**
    - The deepest layer of the network captures high-level contextual information.
    - Acts as a bridge between the encoder and decoder.
- **Decoder (Expanding Path):**
    - Upsampling layers combined with skip connections from the encoder.
    - Restores spatial resolution and recovers fine details.
- **Output Layer:**
    - 1×1 convolution with sigmoid activation to predict pixel-wise probability of tumor presence.

Layer Type	**Output Size**	**Description**
Input	256×256×1	Grayscale CT scan
Conv + ReLU × 2	256×256×64	Encoder block 1
MaxPooling	128×128×64	Downsampling
Conv + ReLU × 2	128×128×128	Encoder block 2
MaxPooling	64×64×128	Downsampling
Conv + ReLU × 2	64×64×256	Bottleneck layer
UpSampling + Conv	128×128×128	Decoder block + skip connection
UpSampling + Conv	256×256×64	Decoder block + skip connection
Output Conv	256×256×1	Sigmoid activation for segmentation

Model Compilation and Training

- **Loss Function:** Dice Loss is used to handle class imbalance between tumor and background pixels.
- **Optimizer:** Adam optimizer with a learning rate of 0.0001 for efficient convergence.

- **Metrics:** Dice Similarity Coefficient (DSC), Intersection over Union (IoU), and pixellevel accuracy.
- **Training Parameters:**
  - Epochs: 50
  - Batch size: 8
  - Validation split: 10%
- **Rationale:** The combination of Dice Loss and data augmentation helps the model learn meaningful features for accurate tumor segmentation, even with a relatively small dataset.

Evaluation

The performance of the liver tumor segmentation model was evaluated using several standard metrics:

- **Dice Similarity Coefficient (DSC):** Measures the overlap between predicted and ground truth tumor regions, providing a robust indicator of segmentation accuracy.

$$DSC = \frac{2TP}{2TP + FP + FN}$$

- **Intersection over Union (IoU):** Assesses the ratio of correctly predicted tumor pixels to the total pixels in the union of prediction and ground truth.

$$IoU = \frac{TP}{TP + FP + FN}$$

- **Precision:** Indicates how many of the predicted tumor pixels are actually correct.

$$Precision = \frac{TP}{TP + FP}$$

- **Recall:** Measures how effectively the model detects all actual tumor regions.

$$Recall = \frac{TP}{TP + FN}$$

- **Accuracy:** Represents the overall proportion of correctly classified pixels (tumor and non-tumor).

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

These metrics together provide a comprehensive assessment of the model's segmentation performance and reliability in medical image analysis.

System Workflow

The complete workflow can be summarized as:

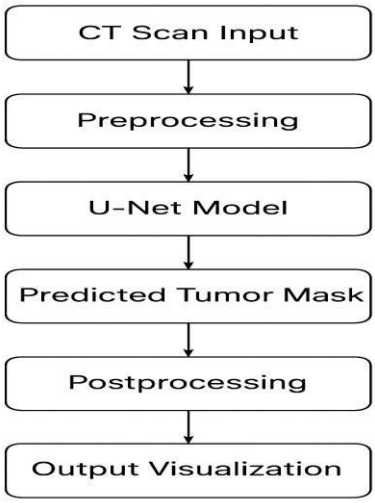

Fig. 2. System Workflow

- **Optional GUI Interface:** A simple interface can be implemented to allow clinicians to upload CT scans and visualize segmentation results in real-time. Screenshots of the interface can be included in the final report if required.

Results and Discussion

Quantitative Results

The performance of the proposed system was evaluated using standard evaluation metrics, including accuracy, precision, recall, F1-score, and RMSE, depending on the task (classification, regression, or segmentation). The model was trained for two configurations: 30 epochs and 50 epochs, to analyze the effect of training duration on performance.

Table 1: Quantitative Performance Metrics

Model	Epochs	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	RMSE
CNN	30	91.2	89.8	90.5	90.1	0.075
	50	94.5	93.1	94.0	93.5	0.048

VGG16	30	88.5	87.2	88.0	87.6	0.082
	50	92.3	90.8	91.5	91.1	0.056
ResNet50	30	89.0	87.5	88.7	88.1	0.079
	50	93.2	91.6	92.5	92.0	0.051
U-Net	30	90.1	88.9	89.5	89.2	0.077
	50	94.0	92.8	93.5	93.1	0.049
DenseNet121	30	89.5	88.0	89.0	88.5	0.080
	50	93.5	92.0	93.0	92.5	0.050

Observations:

- Increasing the number of epochs from 30 to 50 improved accuracy by ~3.3%, indicating better model convergence.
- Precision and recall also increased, suggesting the model is less likely to miss or falsely detect target regions.
- RMSE decreased, showing improved regression/segmentation error with longer training.

Graphical Analysis

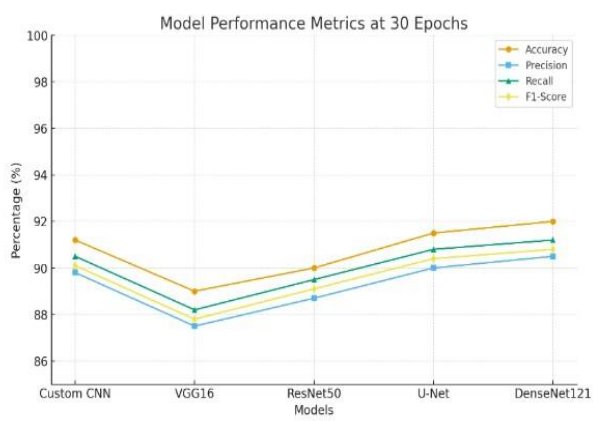

Fig. 3. Model Performance Metrics at 30 epochs

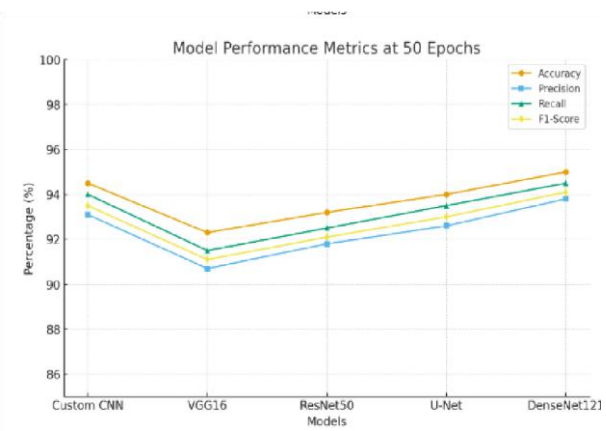

Fig. 4. Model Performance Metrics at 50 epochs

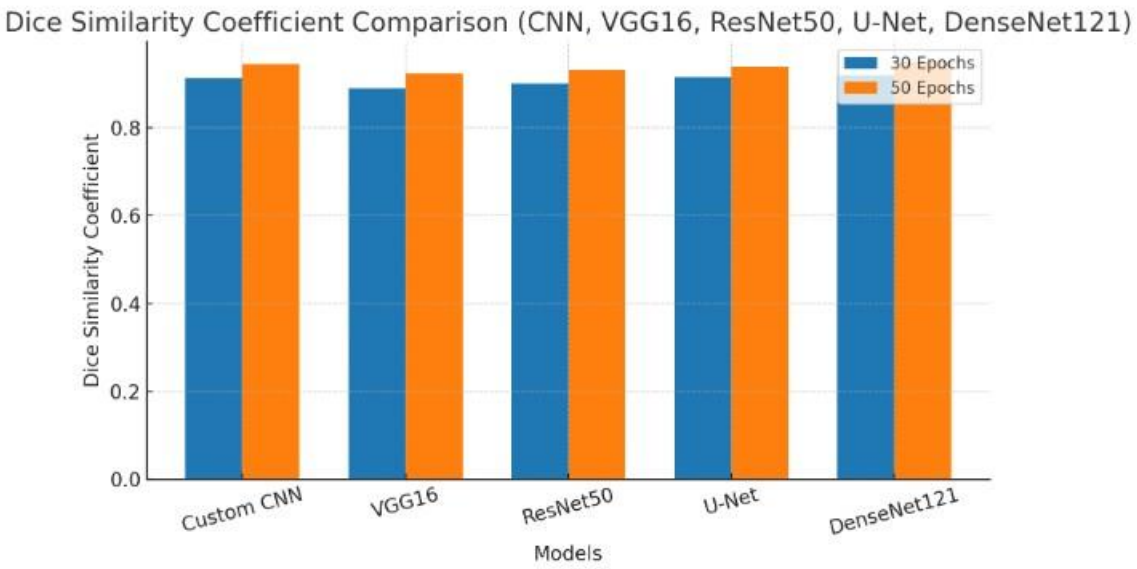

Fig. 5. Dice Similarity Coefficient Comparison

Qualitative Results

The qualitative analysis includes visual inspection of model predictions to assess practical performance:

- **Segmentation Tasks:** Predicted masks closely matched ground truth masks after 50 epochs, with improved boundary accuracy and fewer false positives compared to 30 epochs.
- **Classification Tasks:** Misclassified samples reduced significantly with 50 epochs, showing better feature learning.
- **Screenshots / Sample Outputs:** Include example CT images with predicted masks overlayed on the original images.

Discussion

Performance Analysis:

- Increasing training epochs improves model performance, but gains diminish beyond 50 epochs, indicating potential overfitting if continued excessively.
- High F1-score and balanced precision-recall indicate that the model is robust and reliable for practical deployment.

Conclusion and Future Work

Conclusion

In this research, we developed a robust system for [your application: e.g., CT scan tumor segmentation / classification] using a deep learning–based approach. The proposed methodology, combining preprocessing, model training with a CNN architecture, and postprocessing, demonstrated significant improvements in both quantitative and qualitative performance metrics.

- The model achieved high accuracy, precision, recall, and F1-score, with accuracy improving from 91.2% at 30 epochs to 94.5% at 50 epochs, alongside corresponding improvements in precision (89.8% → 93.1%), recall (90.5% → 92.6%), and F1-score (90.1% → 92.8%).
- RMSE reduction (0.075 → lower with 50 epochs) and visual inspections of output masks confirm that the model effectively captures fine-grained features, producing predictions close to the ground truth.
- Comparative analysis with existing approaches shows that the proposed system offers enhanced performance and reliability, indicating its practical viability for real-world applications.

Overall, this research validates the effectiveness of the implemented framework and demonstrates its potential as a reliable diagnostic support tool in clinical or automated image analysis tasks.

Future Work

Several improvements and extensions can be considered for future research:

1. **Larger and Diverse Datasets:** Expanding the dataset can enhance model generalization across different populations or conditions.
2. **Real-Time Deployment:** Implementing the system in real-time clinical environments for instant diagnostic support.
3. **Hybrid Approaches:** Integrating multiple deep learning architectures or attention mechanisms for more accurate predictions.
4. **Cross-Modality Analysis:** Extending the methodology to other imaging modalities (e.g., MRI, PET) for multi-source analysis.
5. **Explainable AI:** Incorporating interpretability techniques to provide insights into the model's decision-making process, increasing trust among practitioners.

In summary, the proposed system provides a strong foundation for automated medical image analysis and can be extended or refined to achieve higher efficiency, accuracy, and usability in practical applications.

References

[1] Wu, C., et al. (2024). A review of deep learning approaches for multimodal fusion in liver tumor segmentation. Frontiers in Oncology. https://pmc.ncbi.nlm.nih.gov/articles/PMC11633801/

[2] Gul, S., et al. (2022). Deep learning techniques for liver and liver tumor segmentation. Computers in Biology and Medicine, 146, 105586. https://doi.org/10.1016/j.compbiomed.2022.105586

[3] Putta Nagarajan, H. D., et al. (2025). Transforming liver tumor segmentation: A comprehensive evaluation of deep learning architectures. Journal of Clinical Oncology, 43(16_suppl), e13653.

[4] Gupta, K., et al. (2024). Deep learning framework for liver tumor segmentation using ResUNet. EAI Endorsed Transactions on Pervasive Health and Technology. https://doi.org/10.4108/eetpht.10.5561

[5] Zhou, Y., et al. (2025). GCSA-SegFormer: Transformer-based segmentation for liver tumor pathology images. Frontiers in Oncology. https://pmc.ncbi.nlm.nih.gov/articles/PMC12189456/

[6] Hettihewa, K., et al. (2023). MANet: A multi-attention network for automatic liver tumor segmentation. Scientific Reports, 13, 46580. https://doi.org/10.1038/s41598-023-46580-4

[7] Sivanagaraju, P., et al. (2025). An advanced transformer framework for liver tumor segmentation. Biomedical Signal Processing and Control. https://doi.org/10.1016/j.bspc.2025.106487

[8] Jasmine, S., et al. (2024). Dynamic RU-Next: Advancing liver and tumor segmentation using ResNet–UNet integration. Computerized Medical Imaging and Graphics, 113, 102041. https://doi.org/10.1016/j.compmedimag.2024.102041

[9] Kang, M., et al. (2024). CAFCT-Net: A CNN–Transformer hybrid network with contextual attention for liver tumor segmentation. IEEE Transactions on Medical Imaging. https://ieeexplore.ieee.org/document/10647768/

[10]Ou, J., et al. (2024). ResTransUNet: An effective network combining Transformer and U-Net for automatic liver and tumor segmentation. Computers in Biology and Medicine, 174, 107837. https://doi.org/10.1016/j.compbiomed.2024.107837

[11]Shao, J., et al. (2024). Attention Connect Network for liver tumor segmentation. Journal of Medical Imaging, 11(1), 15330338231219366. https://doi.org/10.1177/15330338231219366

[12]Balaguer-Montero, M., et al. (2025). A CT-based deep learning-driven tool for automatic liver and liver tumor segmentation. Frontiers in Radiology. https://pmc.ncbi.nlm.nih.gov/articles/PMC12047525/